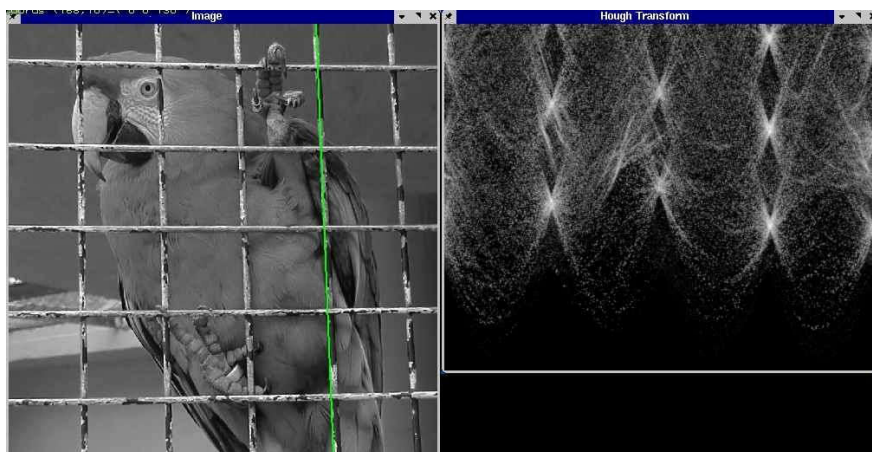

Fitting

(Hough transform)

Lu Peng
School of Computer Science,
Beijing University of Posts and Telecommunications

0

Fitting: The Hough transform



Source: S. Lazebnik

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Voting schemes

- Let each feature vote for all the models that are compatible with it
- Hopefully the noise features will not vote consistently for any single model
- Missing data doesn't matter as long as there are enough features remaining to agree on a good model

Source: S. Lazebnik

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Hough transform

- An early type of voting scheme
- General outline:
 - Discretize parameter space into bins
 - For each feature point in the image, put a vote in every bin in the parameter space that could have generated this point
 - Find bins that have the most votes

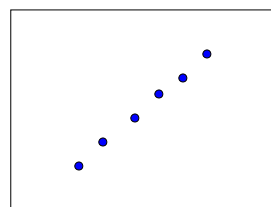
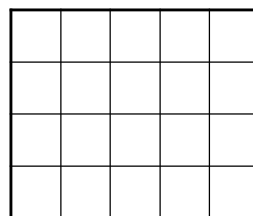


Image space



Hough parameter space

P.V.C. Hough, *Machine Analysis of Bubble Chamber Pictures*, Proc. Int. Conf. High Energy Accelerators and Instrumentation, 1959

Source: S. Lazebnik

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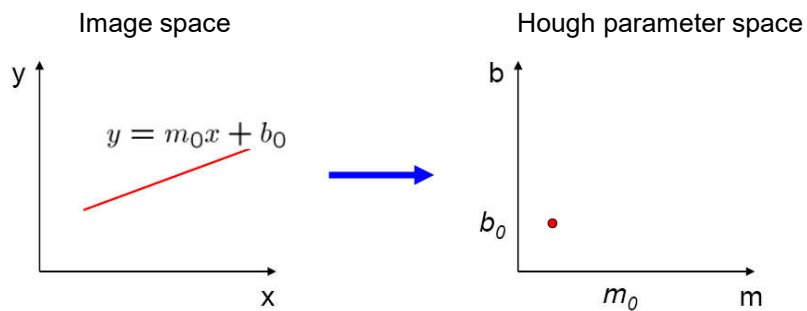
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Parameter space representation

- A line in the image corresponds to a point in Hough space



Source: S. Seitz

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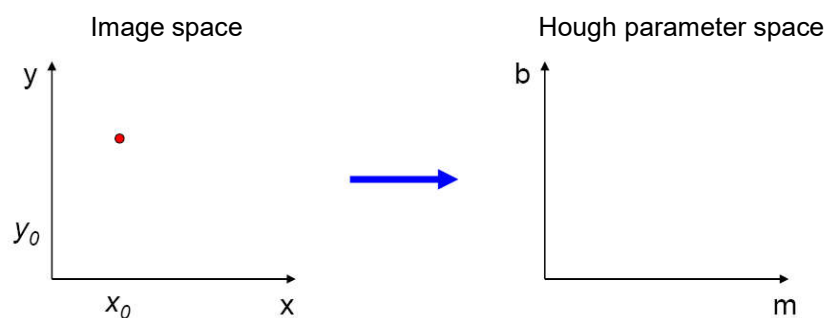
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Parameter space representation

- What does a point (x_0, y_0) in the image space map to in the Hough space?



Source: S. Lazebnik

2020/3/7

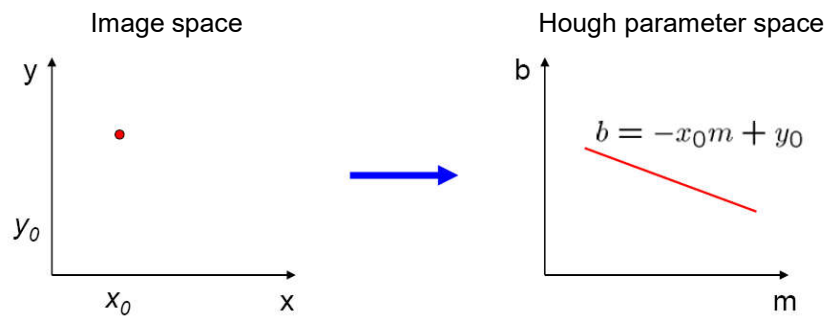
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Parameter space representation

- What does a point (x_0, y_0) in the image space map to in the Hough space?
 - Answer: the solutions of $b = -x_0m + y_0$
 - This is a line in Hough space



Source: S. Lazebnik

2020/3/7

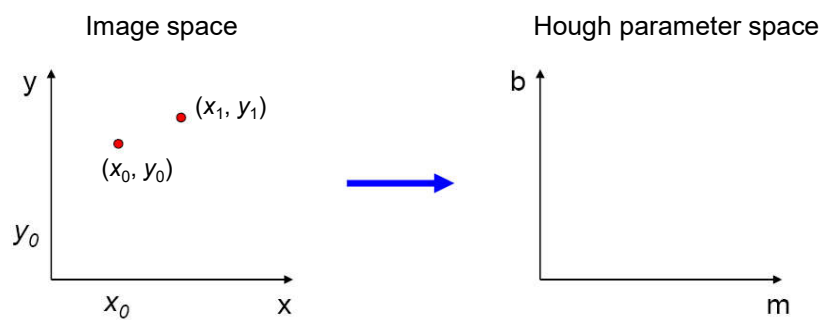
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Parameter space representation

- Where is the line that contains both (x_0, y_0) and (x_1, y_1) ?



Source: S. Lazebnik

2020/3/7

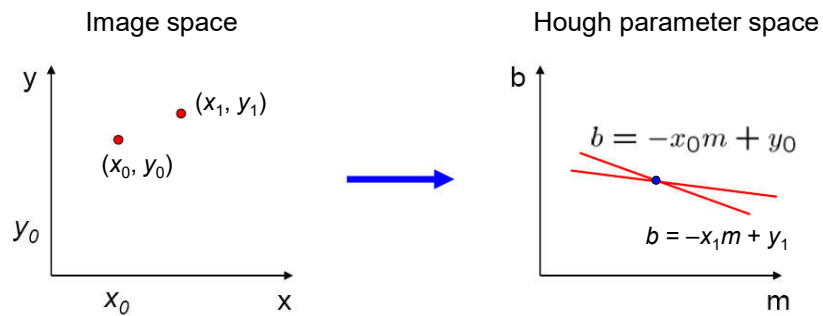
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Parameter space representation

- Where is the line that contains both (x_0, y_0) and (x_1, y_1) ?
 - It is the intersection of the lines $b = -x_0m + y_0$ and $b = -x_1m + y_1$



Source: S. Lazebnik

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Parameter space representation

- Problems with the (m, b) space:
 - Unbounded parameter domain
 - Vertical lines require infinite m

Source: S. Lazebnik

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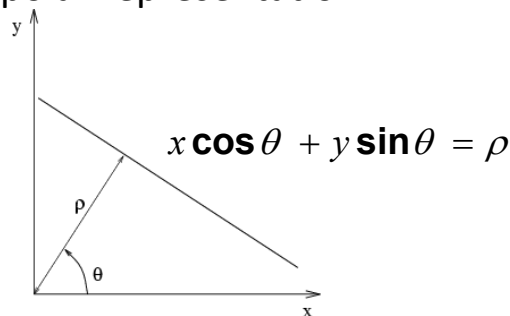
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Parameter space representation

- Problems with the (m,b) space:
 - Unbounded parameter domain
 - Vertical lines require infinite m
- Alternative: polar representation



Each point will add a sinusoid in the (θ, ρ) parameter space

Source: S. Lazebnik

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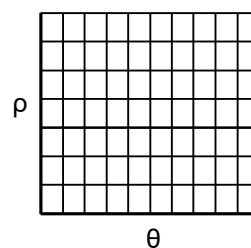
10

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Algorithm outline

- Initialize accumulator H to all zeros
- For each edge point (x,y) in the image
 - For $\theta = 0$ to 180
 - $\rho = x \cos \theta + y \sin \theta$
 - $H(\theta, \rho) = H(\theta, \rho) + 1$
 - end
- Find the value(s) of (θ, ρ) where $H(\theta, \rho)$ is a local maximum
 - The detected line in the image is given by $\rho = x \cos \theta + y \sin \theta$

H: accumulator array (votes)



Source: S. Lazebnik

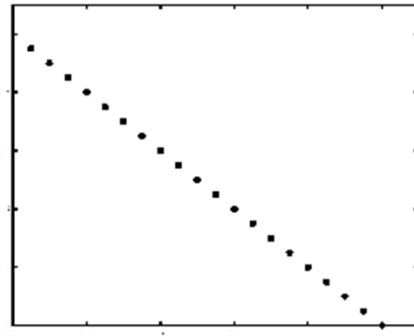
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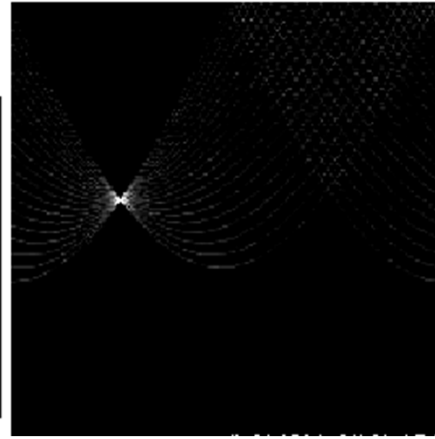
11

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Basic illustration



features



votes

Source: S. Lazebnik

2020/3/7

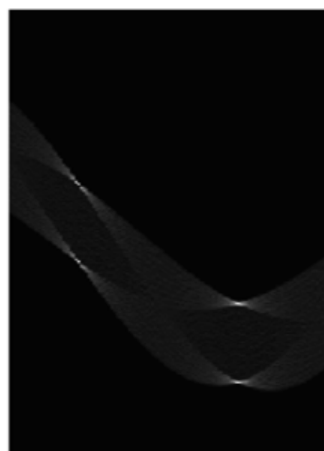
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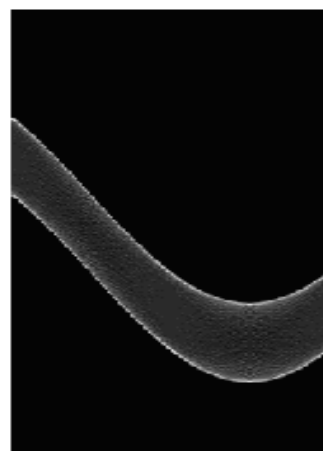
12

Other shapes

Square



Circle



Source: S. Lazebnik

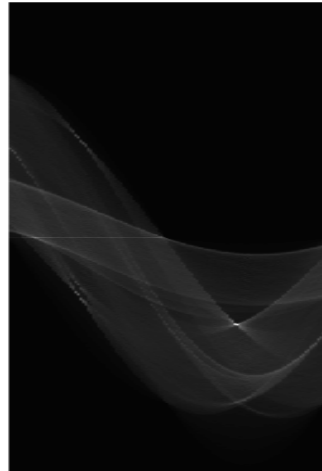
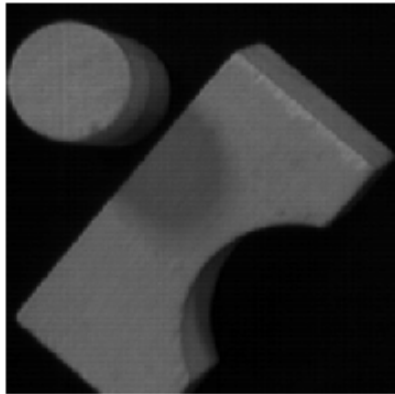
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Several lines



Source: S. Lazebnik

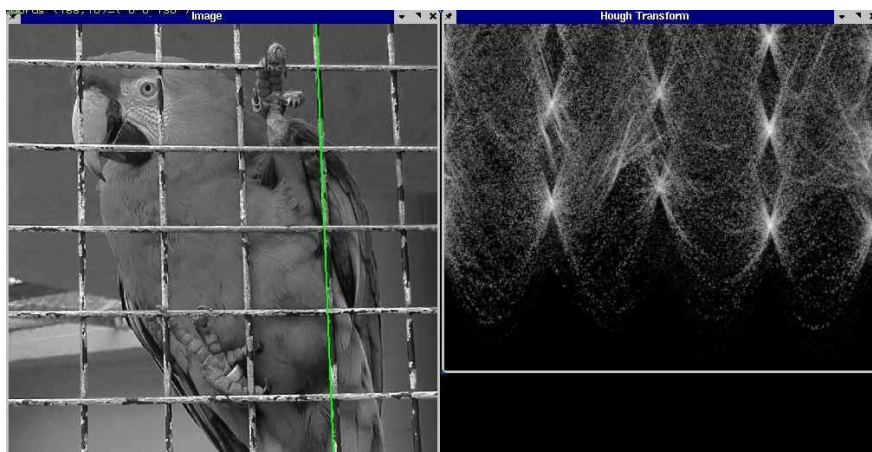
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A more complicated image



http://ostatic.com/files/images/ss_hough.jpg

Source: S. Lazebnik

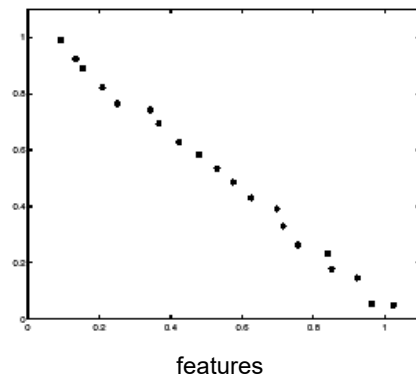
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Effect of noise



Source: S. Lazebnik

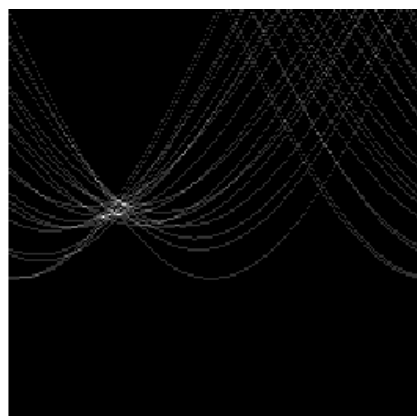
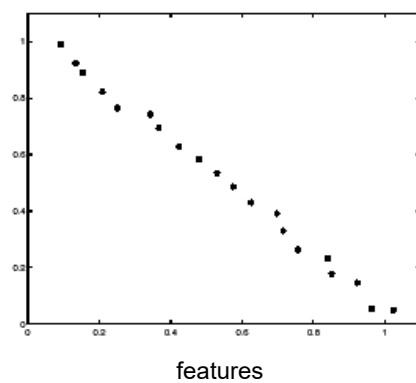
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Effect of noise



Peak gets fuzzy and hard to locate

Source: S. Lazebnik

2020/3/7

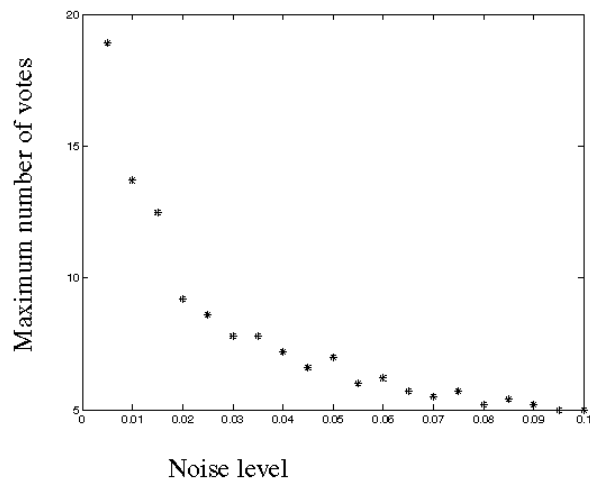
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Effect of noise

- Number of votes for a line of 20 points with increasing noise:



Source: S. Lazebnik

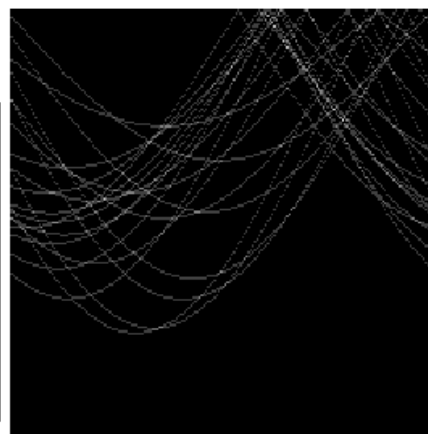
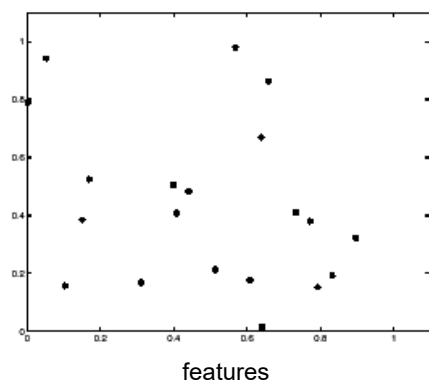
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Random points



Uniform noise can lead to spurious peaks in the array

Source: S. Lazebnik

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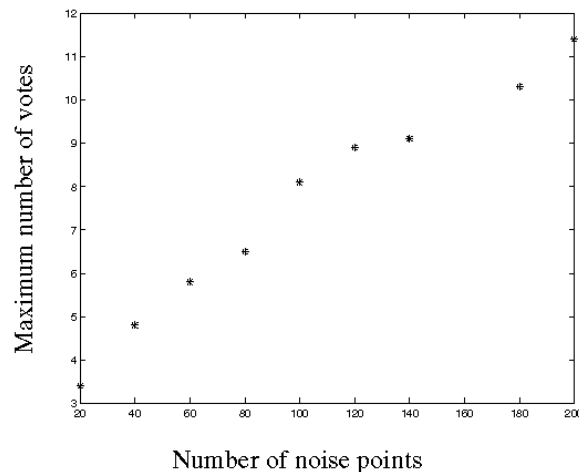
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Random points

- As the level of uniform noise increases, the maximum number of votes increases too:



Source: S. Lazebnik

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Dealing with noise

- Choose a good grid / discretization
 - Too coarse: large votes obtained when too many different lines correspond to a single bucket
 - Too fine: miss lines because some points that are not exactly collinear cast votes for different buckets
- Increment neighboring bins (smoothing in accumulator array)
- Try to get rid of irrelevant features
 - Take only edge points with significant gradient magnitude

Source: S. Lazebnik

2020/3/7

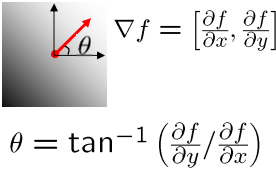
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Incorporating image gradients

- Recall: when we detect an edge point, we also know its gradient direction
- But this means that the line is uniquely determined!
- Modified Hough transform:


$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$
$$\theta = \tan^{-1} \left(\frac{\frac{\partial f}{\partial y}}{\frac{\partial f}{\partial x}} \right)$$

For each edge point (x,y)
 θ = gradient orientation at (x,y)
 $\rho = x \cos \theta + y \sin \theta$
 $H(\theta, \rho) = H(\theta, \rho) + 1$
end

Source: S. Lazebnik

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Hough transform for circles

- How many dimensions will the parameter space have?
- Given an oriented edge point, what are all possible bins that it can vote for?

Source: S. Lazebnik

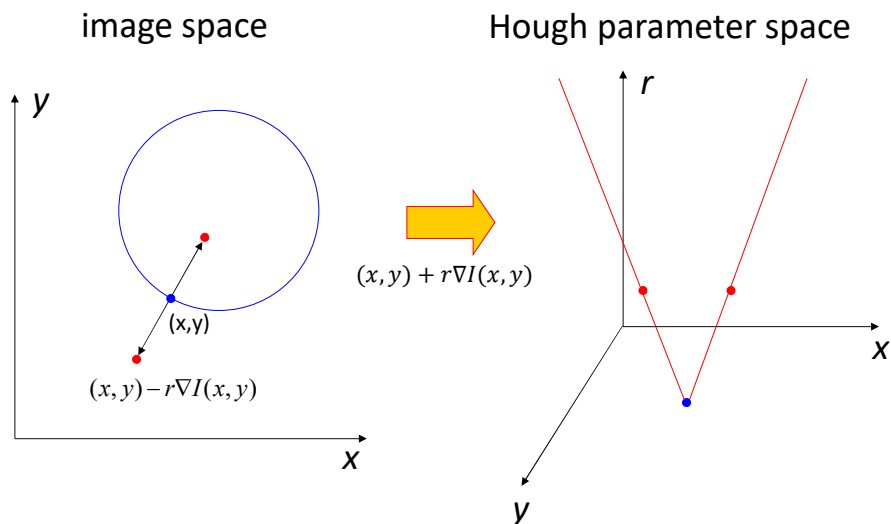
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Hough transform for circles



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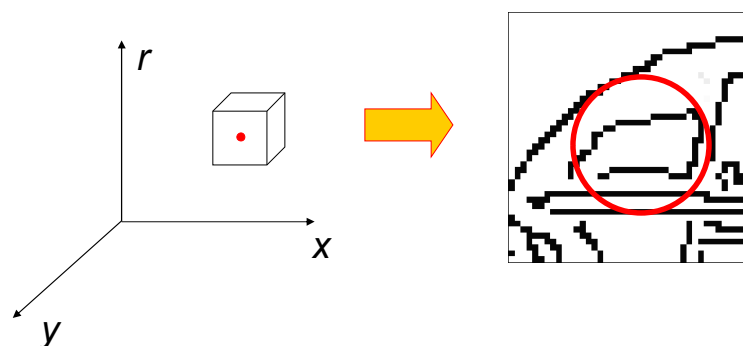
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Hough transform for circles

- Conceptually equivalent procedure: for each (x, y, r) , draw the corresponding circle in the image and compute its "support"



Is this more or less efficient than voting with features?

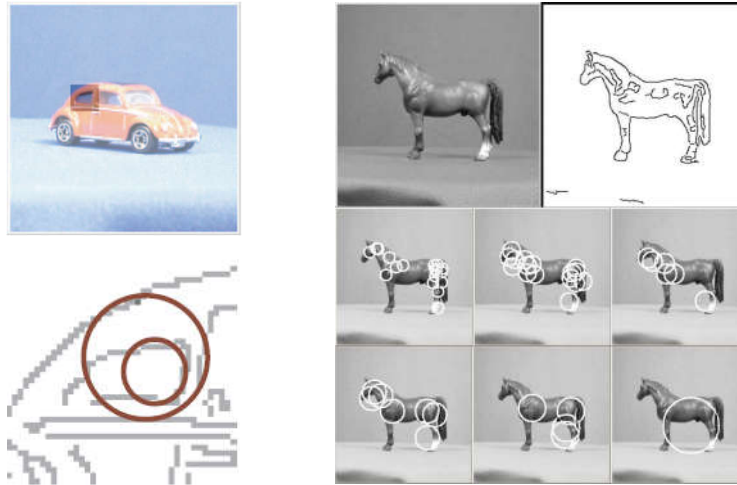
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Application in recognition



F. Jurie and C. Schmid, [Scale-invariant shape features for recognition of object categories](#), CVPR 2004

Source: S. Lazebnik

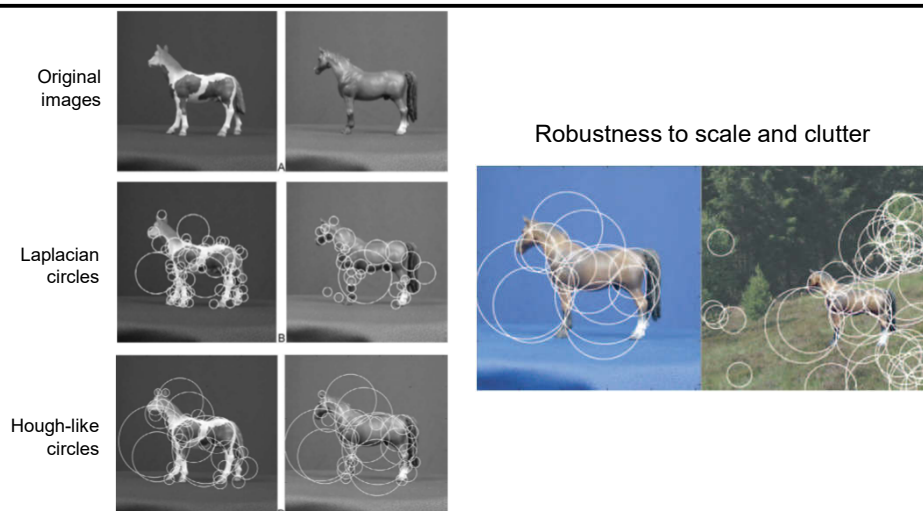
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Hough circles vs. Laplacian blobs



F. Jurie and C. Schmid, [Scale-invariant shape features for recognition of object categories](#), CVPR 2004

Source: S. Lazebnik

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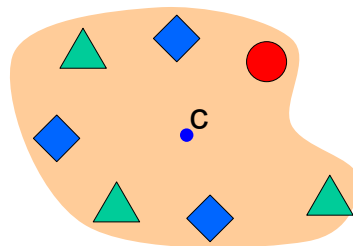
27

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Generalized Hough transform

- We want to find a template defined by its reference point (center) and several distinct types of landmark points in stable spatial configuration

Template



Source: S. Lazebnik

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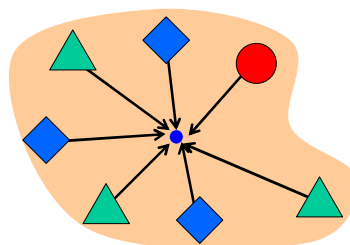
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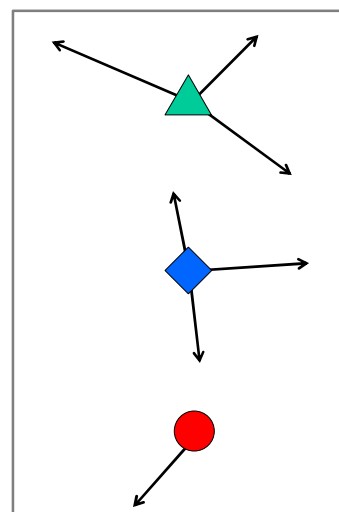
Generalized Hough transform

- Template representation: for each type of landmark point, store all possible displacement vectors towards the center

Template



Model



Source: S. Lazebnik

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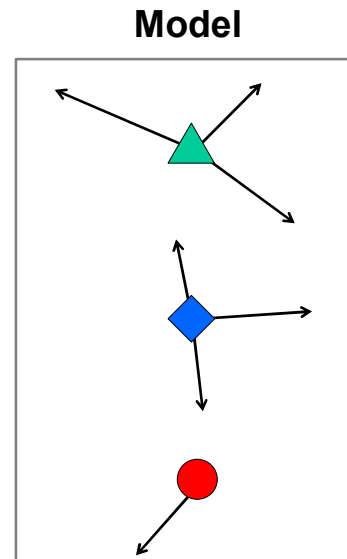
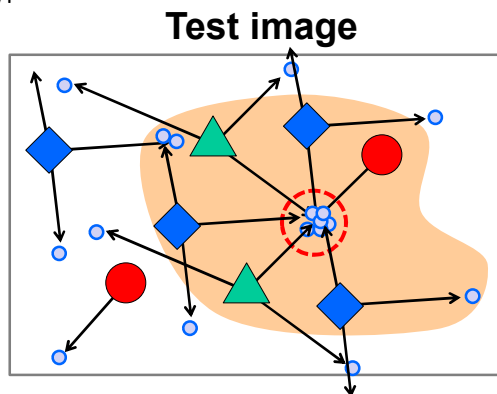
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Generalized Hough transform

- Detecting the template:
 - For each feature in a new image, look up that feature type in the model and vote for the possible center locations associated with that type in the model



Source: S. Lazebnik

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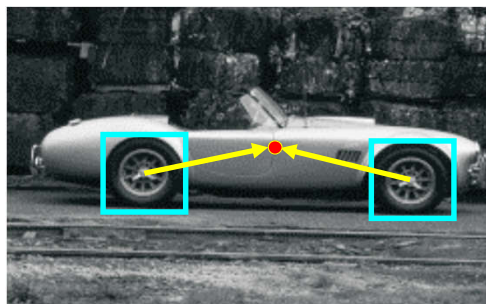
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Application in recognition

- Index displacements by “visual codeword”



training image



visual codeword with displacement vectors

B. Leibe, A. Leonardis, and B. Schiele, [Combined Object Categorization and Segmentation with an Implicit Shape Model](#), ECCV Workshop on Statistical Learning in Computer Vision 2004

Source: S. Lazebnik

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Application in recognition

- Index displacements by “visual codeword”



test image

B. Leibe, A. Leonardis, and B. Schiele, [Combined Object Categorization and Segmentation with an Implicit Shape Model](#), ECCV Workshop on Statistical Learning in Computer Vision 2004

Source: S. Lazebnik

2020/3/7

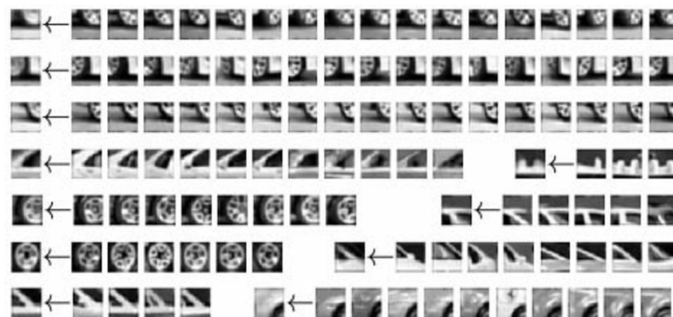
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Implicit shape models: Training

1. Build codebook of patches around extracted interest points using clustering (more on this later in the course)



Source: S. Lazebnik

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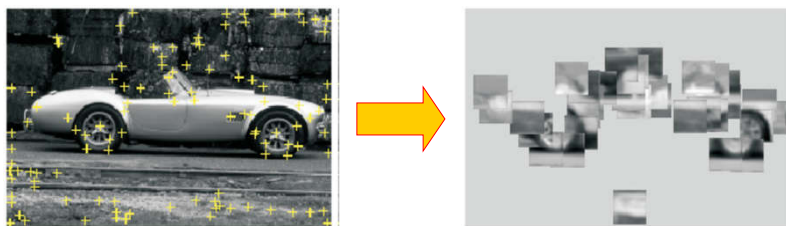
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Implicit shape models: Training

1. Build codebook of patches around extracted interest points using clustering
2. Map the patch around each interest point to closest codebook entry



Source : S. Lazechnik

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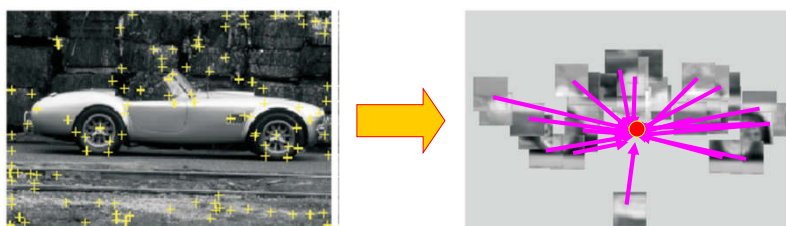
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Implicit shape models: Training

1. Build codebook of patches around extracted interest points using clustering
2. Map the patch around each interest point to closest codebook entry
3. For each codebook entry, store all positions it was found, relative to object center



Source : S. Lazechnik

2020/3/7

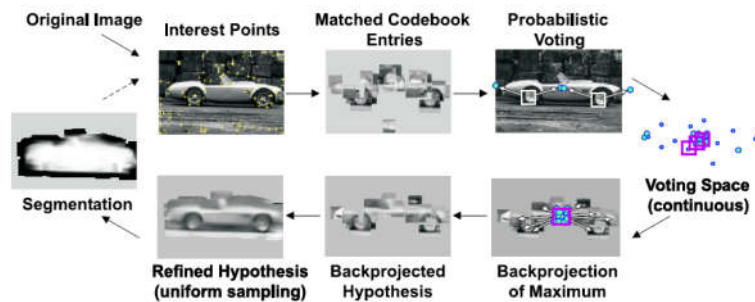
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Implicit shape models: Testing

1. Given test image, extract patches, match to codebook entry
2. Cast votes for possible positions of object center
3. Search for maxima in voting space
4. Extract weighted segmentation mask based on stored masks for the codebook occurrences



Source: S. Lazebnik

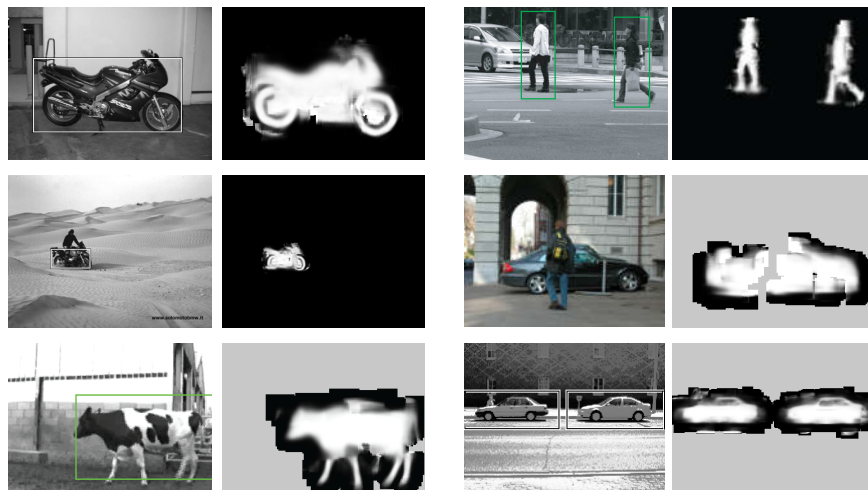
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Additional examples



B. Leibe, A. Leonardis, and B. Schiele, [Combined Object Categorization and Segmentation with an Implicit Shape Model](#), ECCV Workshop on Statistical Learning in Computer Vision 2004

Source: S. Lazebnik

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Implicit shape models: Details

- Supervised training
 - Need reference location and segmentation mask for each training car
- Voting space is continuous, not discrete
 - Clustering algorithm needed to find maxima
- How about dealing with scale changes?
 - Option 1: search a range of scales, as in Hough transform for circles
 - Option 2: use scale-covariant interest points
- Verification stage is very important
 - Once we have a location hypothesis, we can overlay a more detailed template over the image and compare pixel-by-pixel, transfer segmentation masks, etc.

Source: S. Lazebnik

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Hough transform: Discussion

- Pros
 - Can deal with non-locality and occlusion
 - Can detect multiple instances of a model
 - Some robustness to noise: noise points unlikely to contribute consistently to any single bin
- Cons
 - Complexity of search time increases exponentially with the number of model parameters
 - Non-target shapes can produce spurious peaks in parameter space
 - It's hard to pick a good grid size
- Hough transform vs. RANSAC

Source: S. Lazebnik

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